# Generative AI for Neutrino Event in IceCube

## JANUS : Joint generative Ai for Neutrino reconstrUction & Simulation

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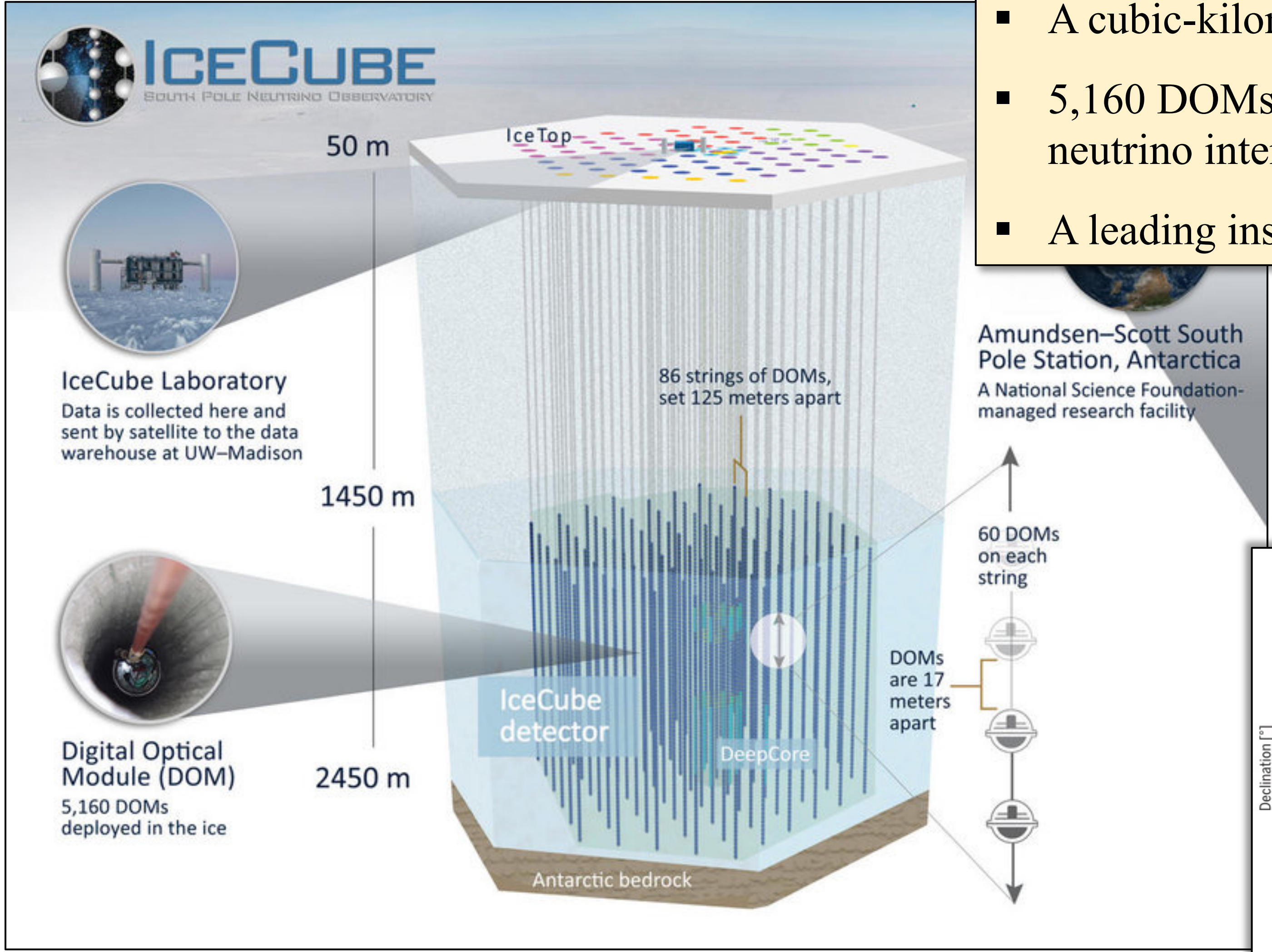
KPS Spring Meeting
Sungkyunkwan University| Department of Physics
Minje Park, Teayun Kim, Chang Dong Rho

# Contents

# About IceCube

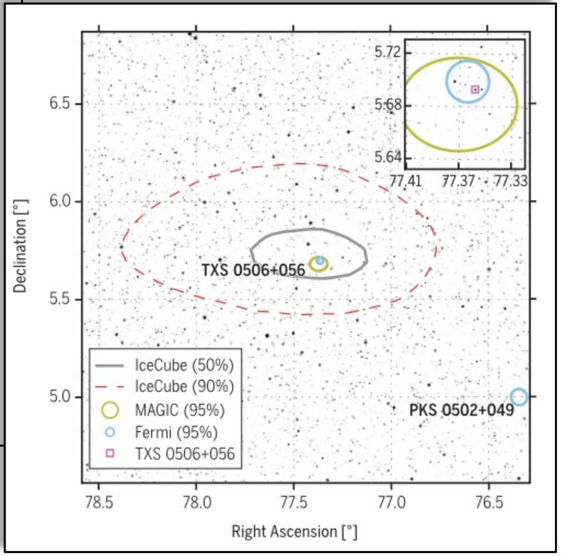


- A cubic-kilometer neutrino detector at the South Pole
- 5,160 DOMs detect Cherenkov light from particles produced by neutrino interactions in ice
- A leading instrument in neutrino astronomy




**Major Achievements**

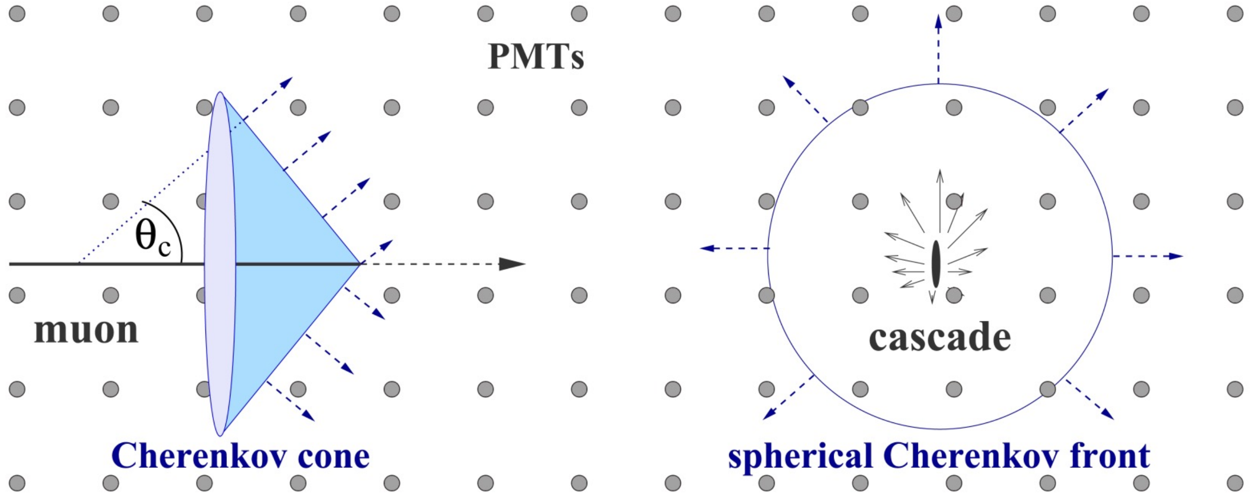
- 2013: Discovery of astrophysical neutrinos
- 2018: First source identified — TXS 0506+056
- 2023: First neutrino map of the Milky Way

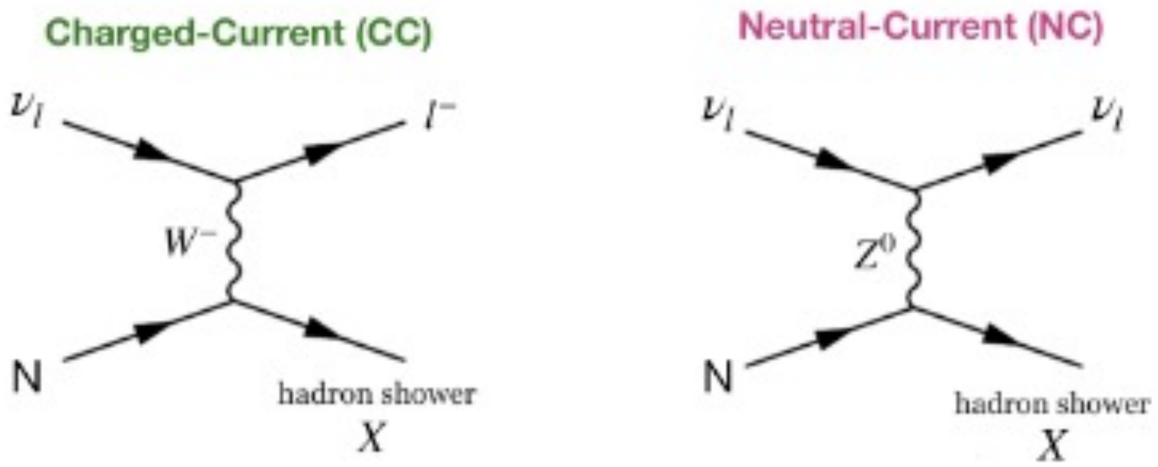



# Neutrino Event Signatures



Neutrino Event in Icecube is divided into 2 different signature **Track signature or Cascade Signature**




When muon neutrino interacts with ice in **CC interaction**,

The resulting muon passes through the detector with **long duration** and **low energy loss**, creating a **long track-like signature**.

All other CC interactions, as well as NC interactions, produce cascade-like event signatures

# Track Reconstruction in IceCube

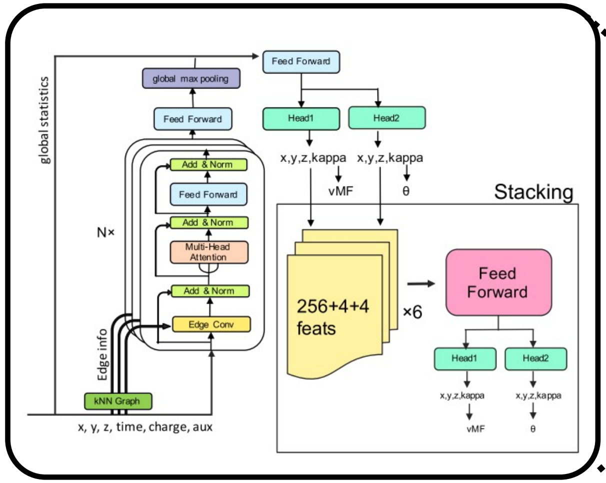


- Track-like events has a angular resolution of $< 1°$ → Main channel for neutrino's source study
- IceCube has mainly two reconstruction methodology for muon reconstruction

**1. Likelihood Reconstruction based on Photon Arrival time Modeling**

**2. Regression Model using Neural Network**

Event




**Neural Network**

**Arrival Time Based Likelihood**

**Line-fit** → **Pandel SPE** → **Pandel MPE** → **Spline MPE**

**Example**

**Many other Network exists**

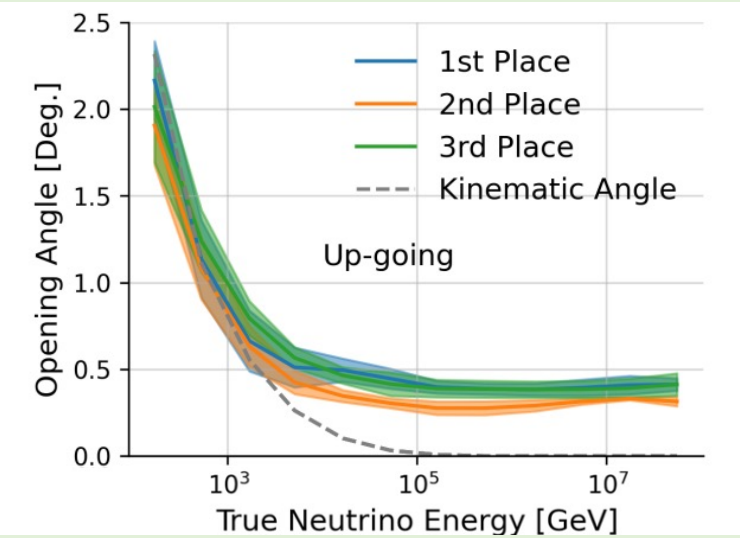
**Track Parameter**

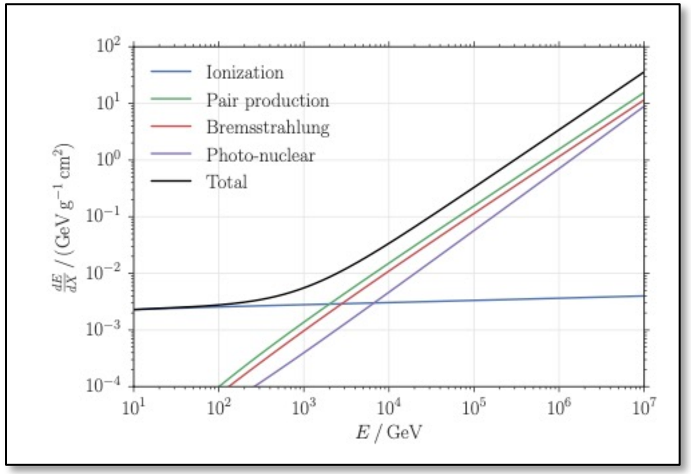
**Example**
**Detail in Appendix**

# Reconstruction Performance & Limitation

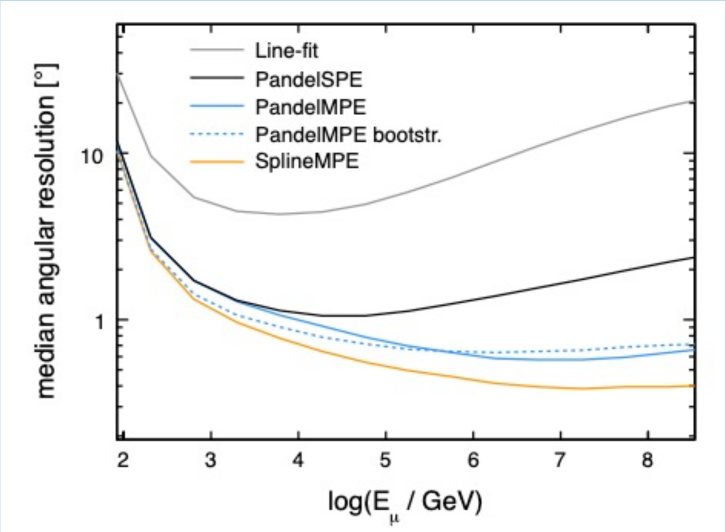


**Neural Network**




**Photon Arrival time based Likelihood**




- **Likelihood-based**: models only photons from muon track → only a fraction of the full event data
- **Neural Network** : relies on spuriously correlated features → poor generalization to out-of-distribution inputs
- **New approach — Generative AI(Flow Matching)**: model event directly & models distribution not the feature

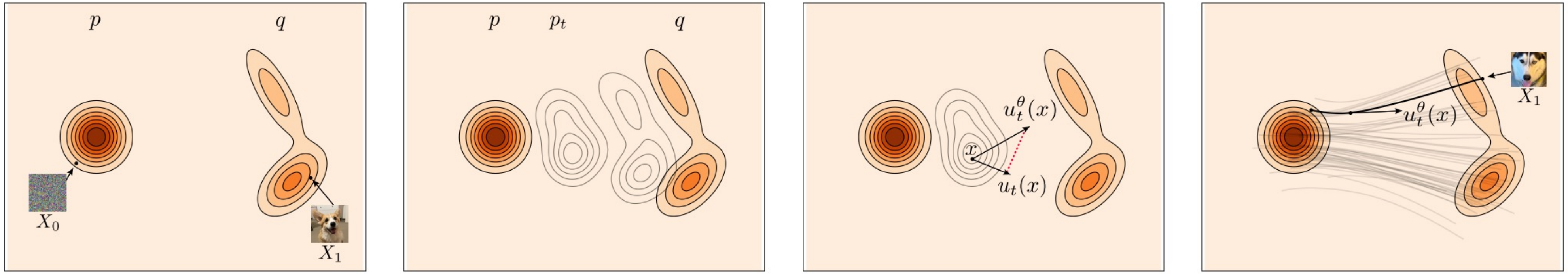
→ Robust Likelihood Reconstruction in high energy muon

# Flow Matching and Generative AI

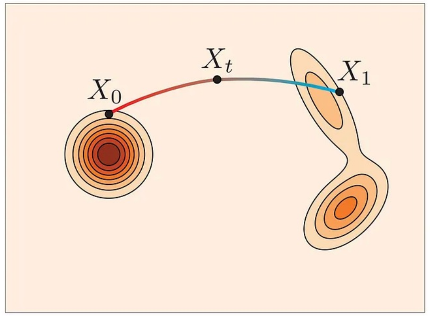


Main objective of modern ***Generative AI*** is to model the data distribution ($q$)

- Start from a simple source distribution $p$ and learn a transformation $p \rightarrow q$ : defined as a **time-dependent path** $p_t$




Data. Path design. Training. Sampling.




**(a)** Flow

## Flow Matching Objective

*flow matching learns a flow $\psi_t$ such that the pushforward of $p_0$ maps to $q$:*

*the velocity field $u_t$ drives the flow $\psi_t$:* $$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x$$

*Training objective becomes find the vector field that satisfy* $$(\psi_1)_{\#}\, p_0 = q$$

$$E[\|u_\theta(x_t, t) - u_t(x_t)\|^2]$$

*pushforward gives exact log-likelihood:* $$\log p_1(x_1) = \log p_0(x_0) - \int_0^1 div\, u_t(x_t)\, dt$$

# JANUS

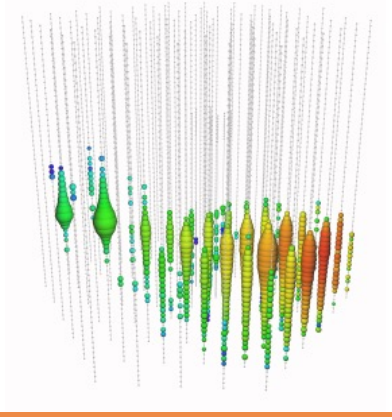


**J**oint generative **A**i for **N**eutrino reconstr**U**ction and **S**imulation

Models is **trained to** generate muon event

*Direction*
*Energy*
*Vertex point*

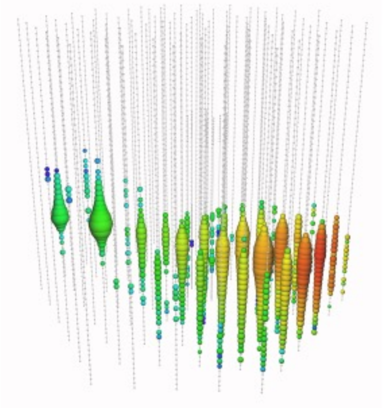
JANUS Model (*Simulator*)



Without additional training model can **infer** physical parameter

*Direction*
*Energy*
*Vertex point*

JANUS Model (*Reconstuctor*)



# Training Objective



- *Original objective of FM is*

$$L_{FM} = E_{\{t,x_0\}} \| u_\theta(x_t, t) - u_t(x_t) \|^2$$

*however, where* $x_1 \sim q_{data}$ *define*

$$L_{CFM} = E_{\{t,x_0,x_1\}} \| u_\theta(x_t, t) - u_t(x_t \mid x_1) \|^2$$

*then*

$$\nabla_\theta L_{FM} = \nabla_\theta L_{CFM}$$

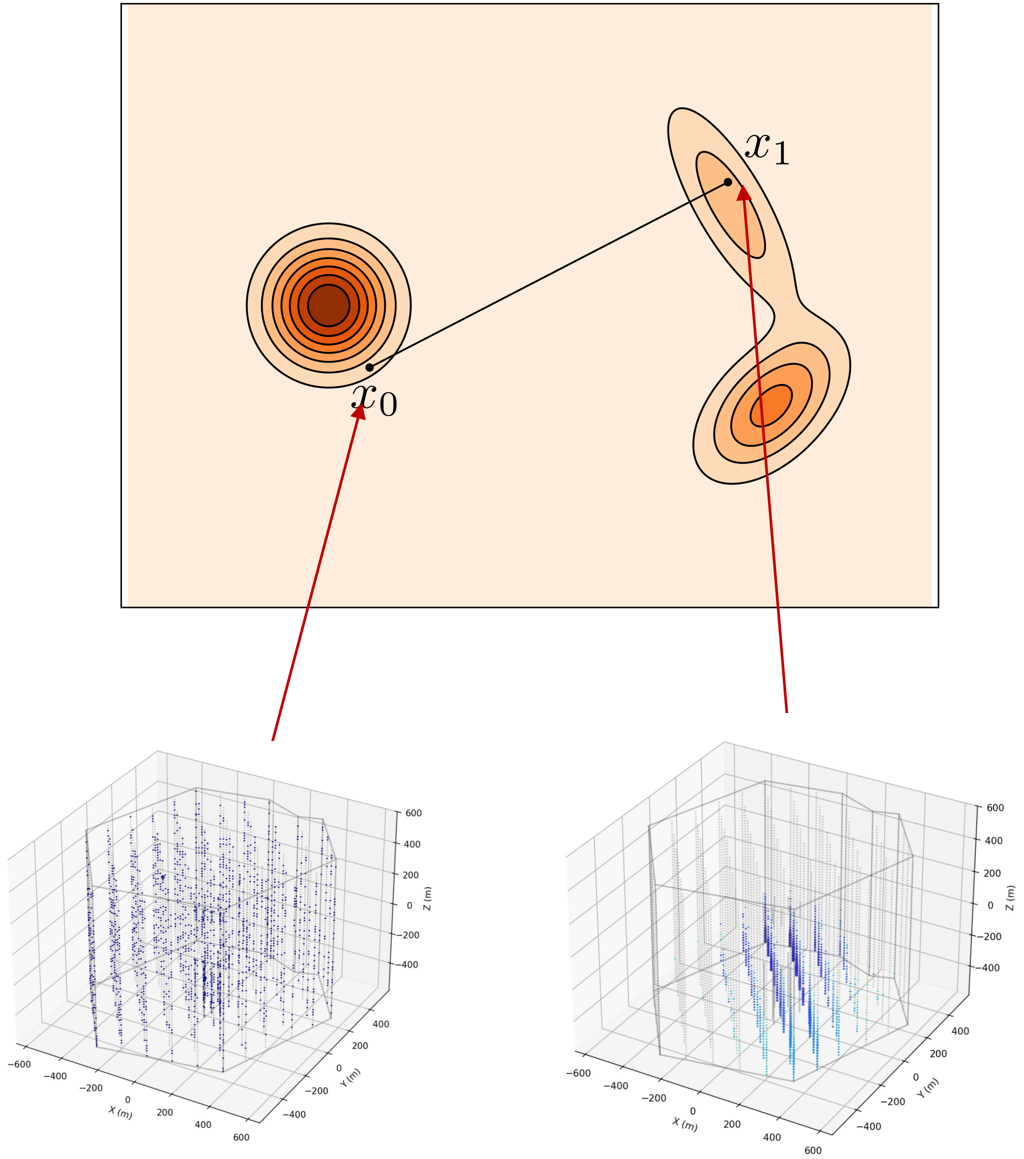
- *Let* $p_t(x_t \mid x_1) = N(tx_1, (1-t)^2 I)$

$$x_t = tx_1 + (1-t)x_0 \sim tx_1 + (1-t)N(0, I)$$

$$\dot{x}_t = x_1 - x_0 = u_t(x_t \mid x_1)$$

- *In conclusion our loss function is in this form*

$$L_{CFM} = E_{\{t,x_0,x_1\}} \| u_\theta(x_t, t) - (x_1 - x_0) \|^2$$



# JANUS Architecture & DataSet

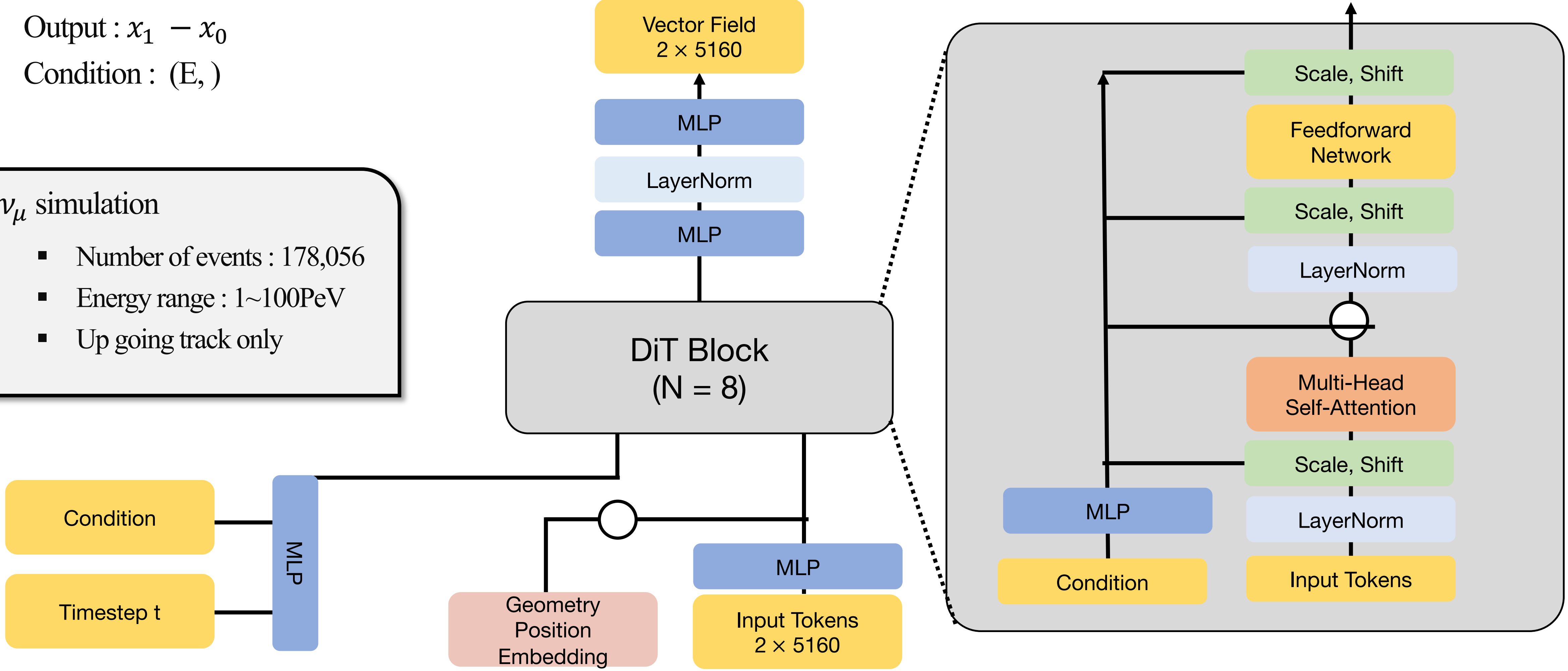


JANUS mainly follow Diffusion Transformer Architecture, with detector geometry position embedding.

- Input : $tx_1 + (1 - t)x_0$ (Vector Field)
- Output : $x_1 - x_0$
- Condition : (E, )

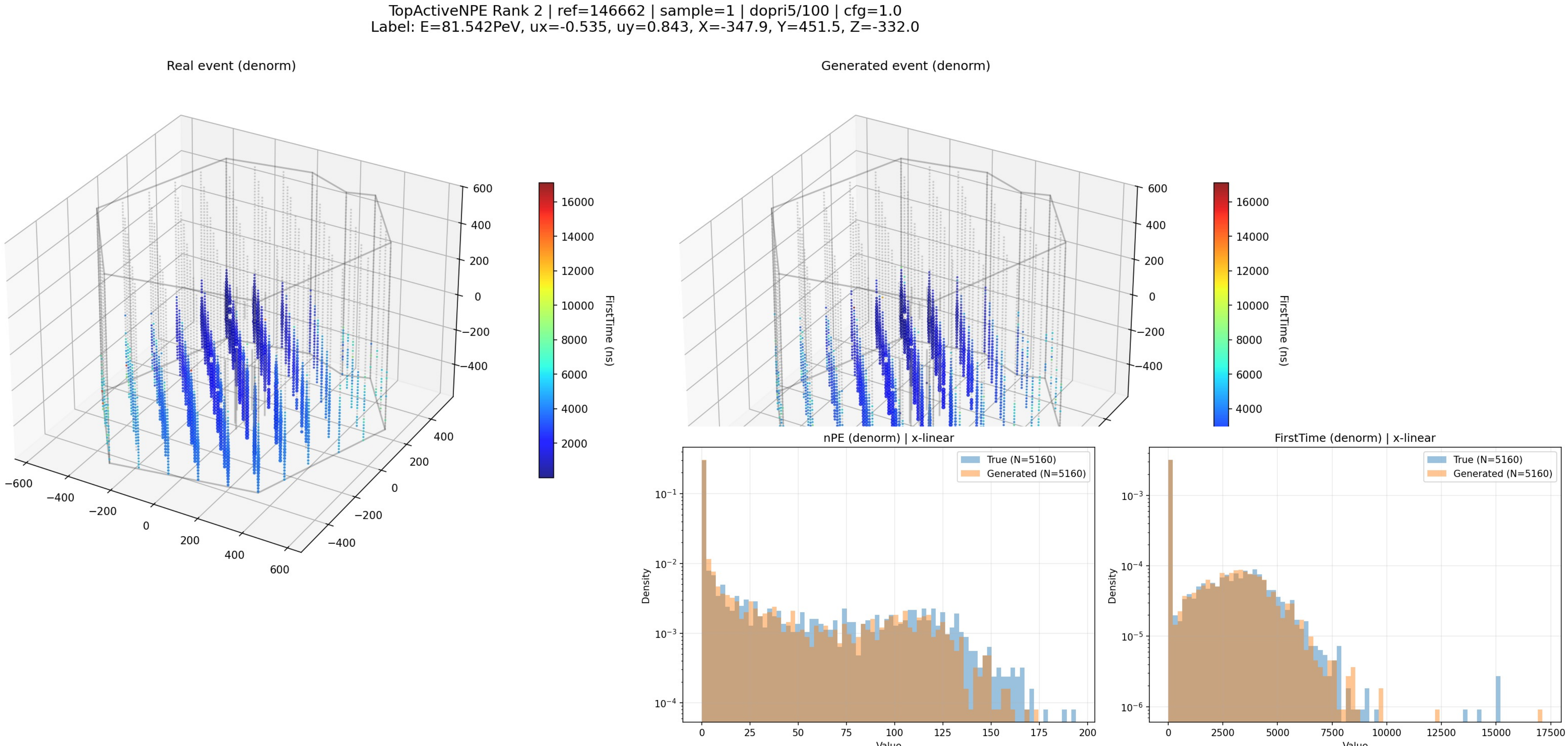
$\nu_\mu$ simulation

- Number of events : 178,056
- Energy range : 1~100PeV
- Up going track only



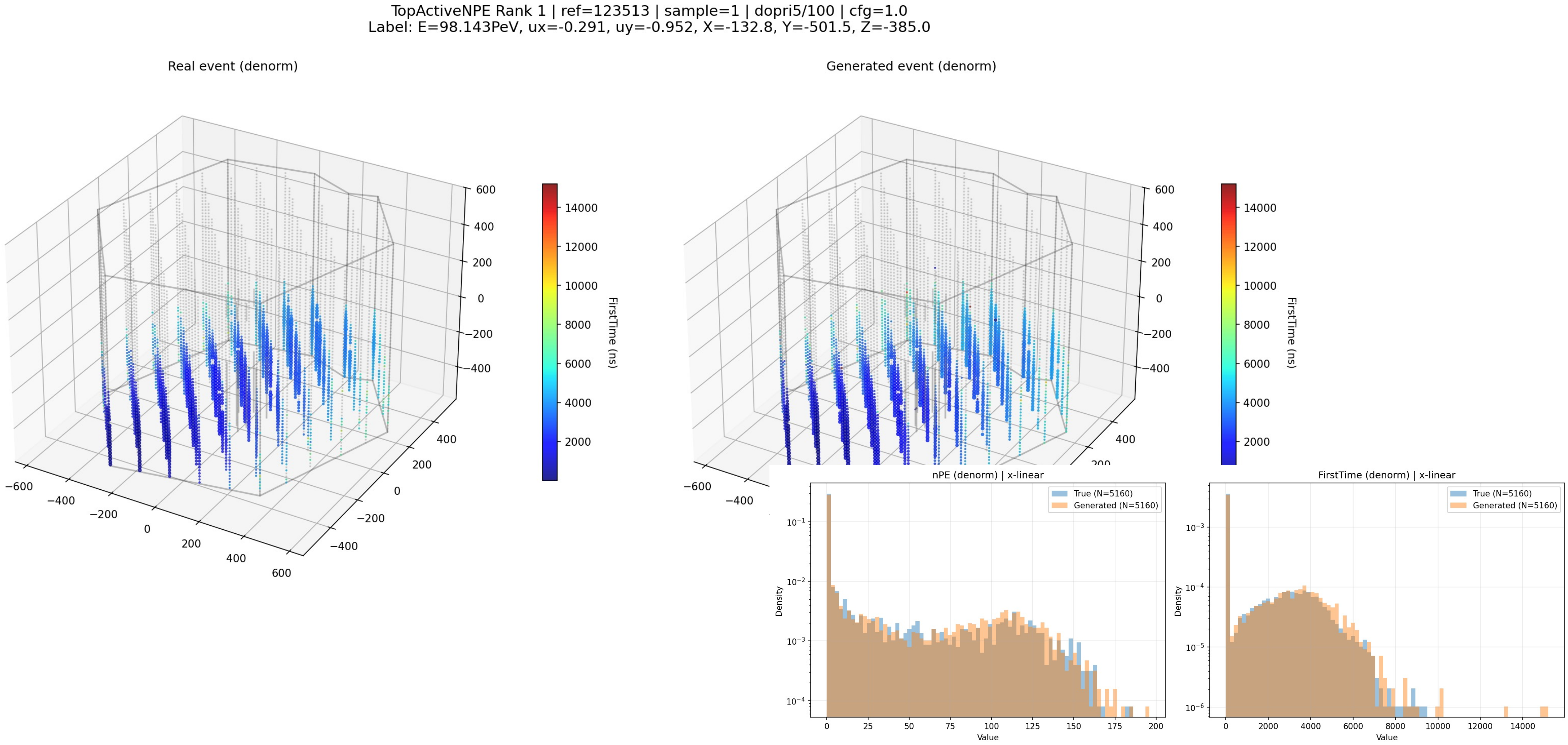
# Sample Performance





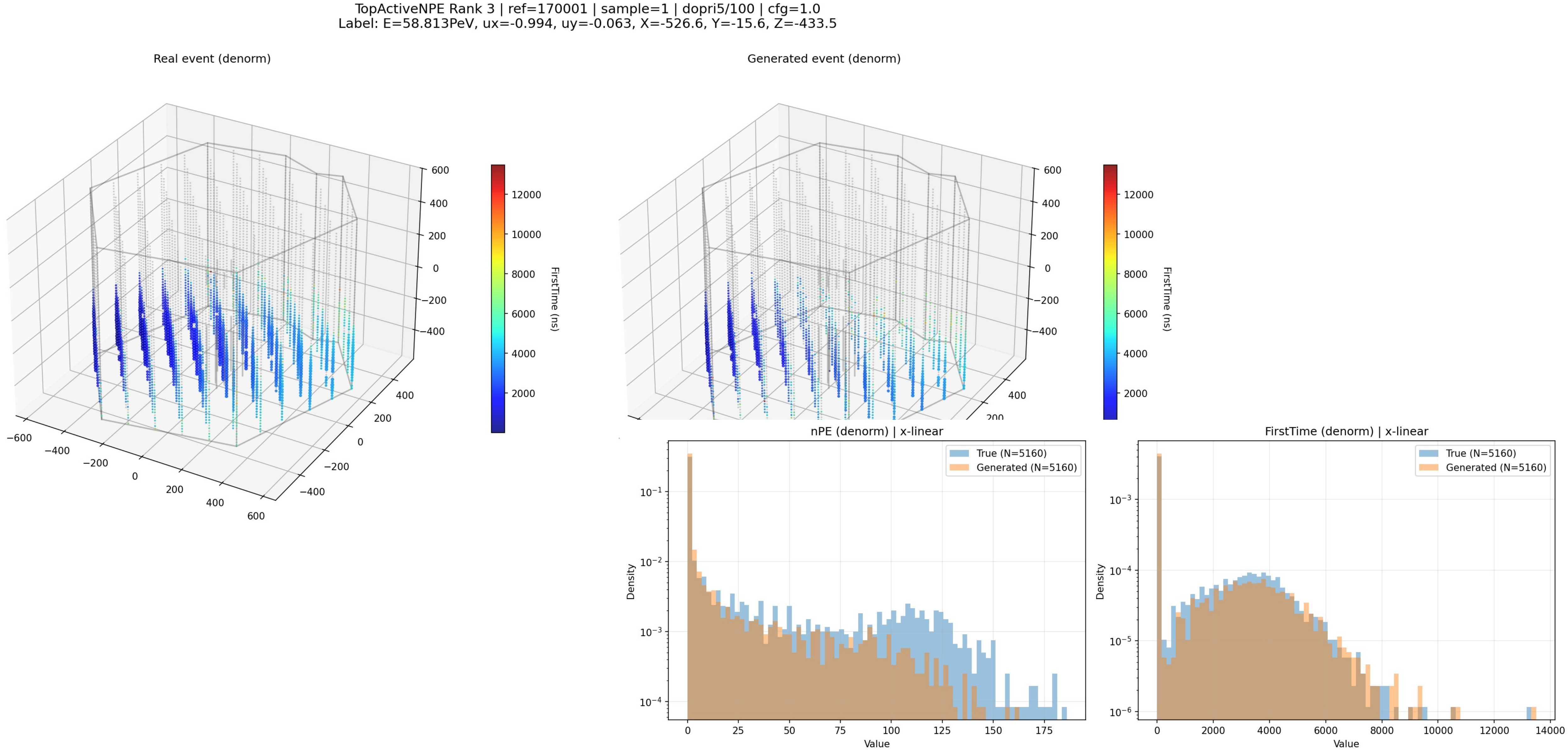
# Sample Performance





# Sample Performance





# Summary & Future Work



- We trained JANUS using a flow matching model
- To the best of our knowledge, this is the first implementation of Generative AI for both event generation
- JANUS appears to capture the overall trends of muon events

# Summary & Future Work



- We trained JANUS using a diffusion model
- To the best of our knowledge, this is the first implementation of Generative AI for both event generation and reconstruction
- JANUS appears to capture the overall trends of muon events
- Likelihood evaluation and parameter inference with JANUS
- Design a summary statistic for the JANUS generated events

# Thank you for your Attention

Fell free to contact me at :

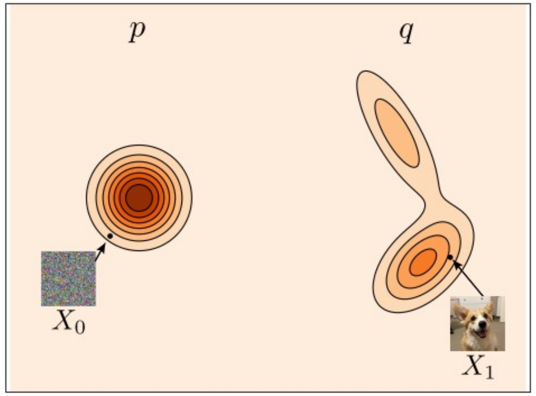
- pmj0324@g.skku.edu
- pmj032400@naver.com


Any questions or discussion is welcome

# How & Why



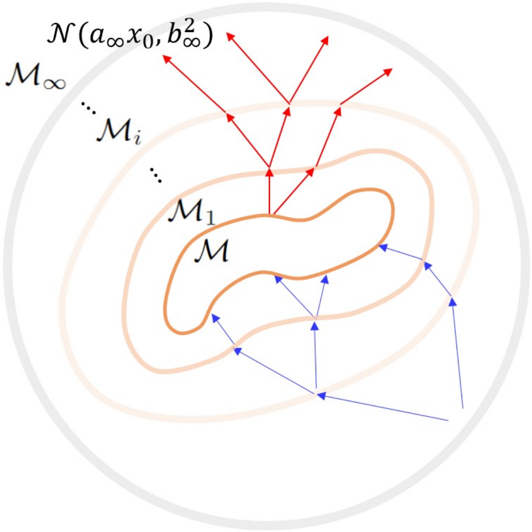
- **How** can we infer from just training to sample the event




(a) Flow
- Deterministic transport (ODE): $\dot{x}_t = v_\theta(x_t, t)$
- $\log p_\theta(x_1 \mid c) = \log p_0(x_0) - \int \nabla_x v_\theta(x_t, t \mid c) dt$

(b) Diffusion
- Stochastic noising / denoising (SDE / reverse process)
- $\log p_\theta(x_1 \mid c) \geq \mathrm{E}_{q(x_{0:1}|x_1)} [\log p_0(x_0) + \int_0^1 \log p_\theta(x_t|x_{t+dt}, c) dt - \int_0^1 \log q(x_t|x_{t+dt}) dt]$

- **Why** should we use it rather than basic regression model

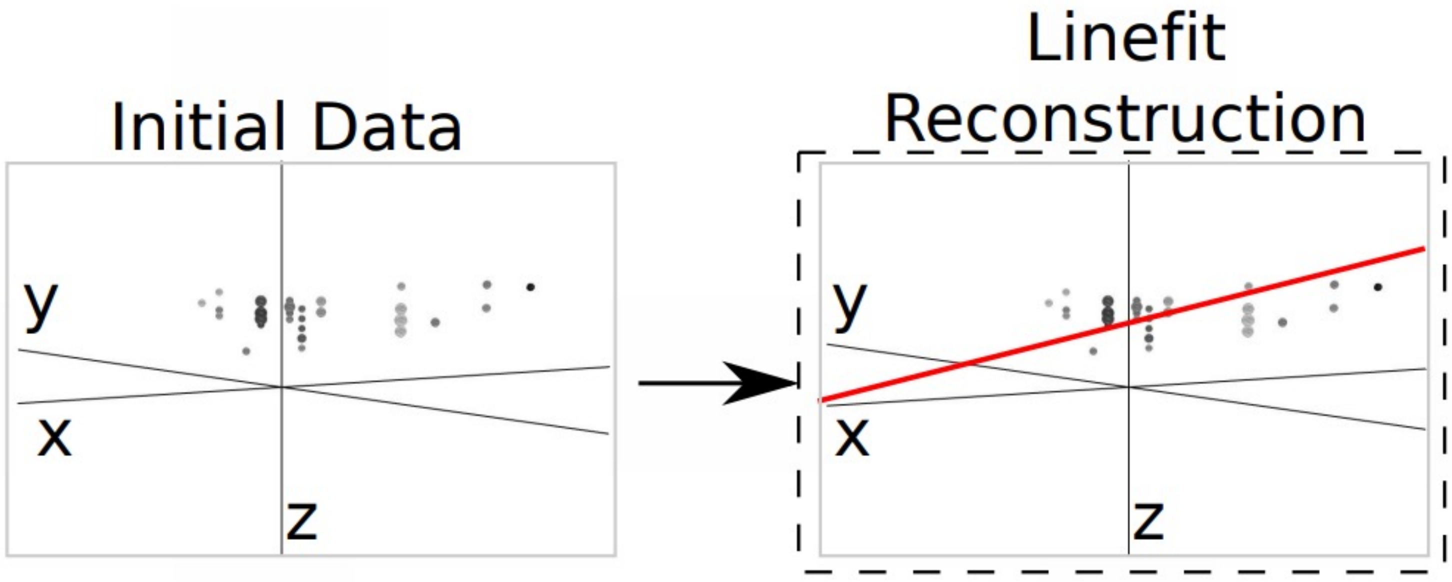



- *Manifold hypothesis*: data lie on a low-dimensional manifold($q$)
- Events from the same underlying muon cluster around the same manifold.
- A generative model learns $q$ itself, enabling manifold-consistent, likelihood-based inference that is more robust to stochastic variability.

# Line-fit



- Very first step is called line-fit
- Simply finding the track that minimizes the squared distance errors between the triggered DOMs




- Obviously optical ice properties are not taken into account

# PandelSPE

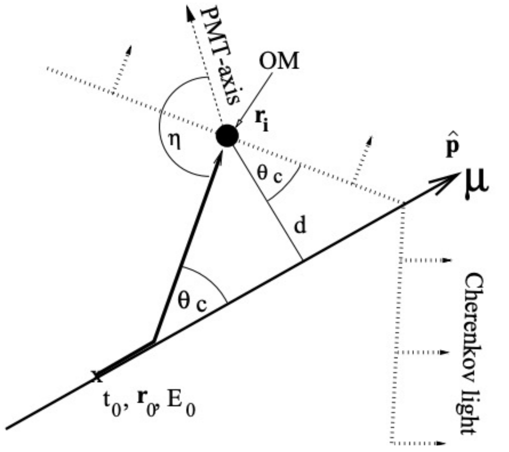


- SPE : Single Photo Electron
- PandelSPE uses the arrival time information of only first Cherenkov photon.

$$\mathcal{L}_{\mathrm{SPE}}(\vec{x}|\vec{\theta}) = \prod_{i}^{\text{1st hits}} \boxed{p(t_{\mathrm{res},i}|\vec{\theta})} \qquad t_{\mathrm{res}} = t_{\mathrm{hit}} - t_{\mathrm{geo}}$$

**Pandel Function**

- Photon arrival time depends on the ice property and relative position between DOM and muon track.
- Studies have motivated the use of the Pandel function to model the photon arrival time




$$t_{\mathrm{geo}} = t_0 + \frac{\hat{\mathbf{p}} \cdot (\mathbf{r_i} - \mathbf{r_0}) + d \tan\theta_c}{c_{\mathrm{vac}}}$$

$$p\left(t_{\mathrm{res}}|\vec{\theta}\right) = \frac{1}{N(d)} \frac{\tau^{-(d/\lambda)} \cdot t_{\mathrm{res}}^{(d/\lambda-1)}}{\Gamma(d/\lambda)} \cdot \exp\left(-t_{\mathrm{res}} \cdot \left(\frac{1}{\tau} + \frac{c_{\mathrm{medium}}}{\lambda_a}\right) - \frac{d}{\lambda_a}\right),$$

$$N(d) = e^{-d/\lambda_a} \cdot \left(1 + \frac{\tau \cdot c_{\mathrm{medium}}}{\lambda_a}\right)^{-d/\lambda},$$

Scattering length , Absorption length, is fixed to the values following

$$\lambda = 33.3\,\mathrm{m} \quad \lambda_a = 98\,\mathrm{m} \quad \tau = 557\,\mathrm{ns}.$$

Only free parameter is ***d***, distance between the track and DOM.

# PandelMPE



- MPE : Single Photo Electron
- PandelMPE uses the first arrival time information and number of measured photon

$$\mathcal{L}_{\mathrm{MPE}}(\vec{x}|\vec{\theta}) = \prod_{i}^{\text{1st hits}} n_i \cdot \boxed{p(t_{\mathrm{res},i}|\vec{\theta})} \cdot (1 - \mathrm{P}(t_{\mathrm{res},i}|\vec{\theta}))^{n_i - 1} \qquad P(t_{\mathrm{res}}|\vec{\theta}) = \int_{-\infty}^{t_{\mathrm{res}}} p(t|\vec{\theta})\mathrm{d}t$$

**Pandel Function**

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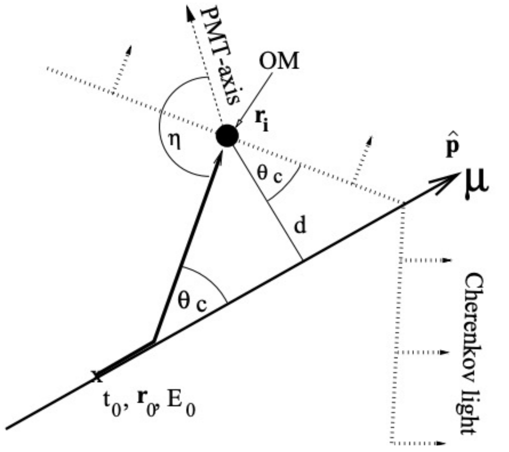


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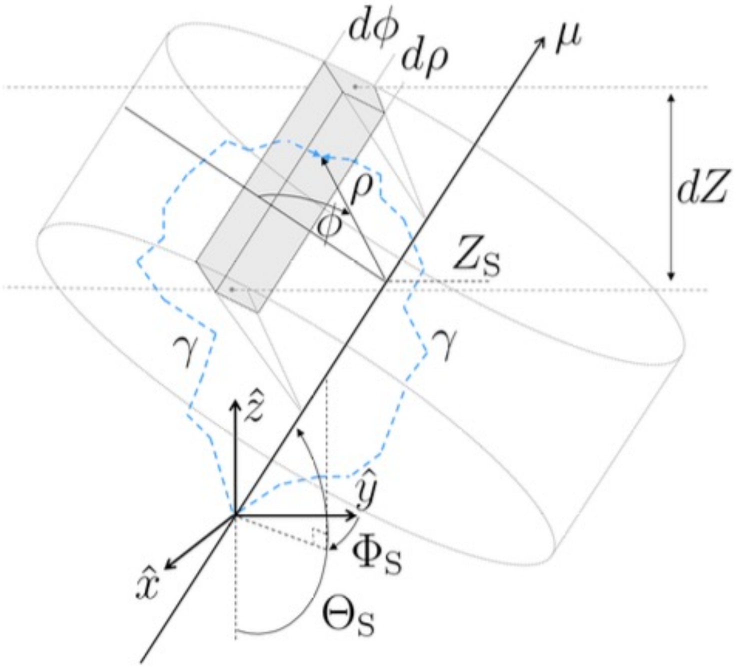
# SPlineMPE



- But we know that scattering & absorption length varies over depth.
- So setting absorption & scattering length fixed , which is done in Pandel MPE, is not great option




| Dimension | # bins per dimension old Photonics | this work | Minimum | Maximum |
|---|---|---|---|---|
| Zenith angle $\Theta_S$ | 18 | 18 | 0° | 180° |
| Depth $Z_S$ | 75 | 75 | −850 m | 650 m |
| Azimuthal $\phi$ | 9 | 36 | 0° | 180° |
| Distance $\rho$ | 30 | 200 | 0 m | 580 m |
| Time residual $t_{res}$ | 50 | 210 | 0 ns | 7000 ns |

**Table 3.1:** Photon table binning

- In this context SplineMPE accrued
  SplineMPE uses likelihood formula same as PandelMPE

$$\mathcal{L}_{\mathrm{MPE}}(\vec{x}|\vec{\theta}) = \prod_{i}^{\text{1st hits}} n_i \cdot p(t_{\mathrm{res},i}|\vec{\theta}) \cdot (1 - \mathrm{P}(t_{\mathrm{res},i}|\vec{\theta}))^{n_i - 1}$$

But Instead of using Pandel function it creates
PDF based on photon distribution table binned as Table 3.1,
Also it uses splines (B-spline) technique for the interpolations

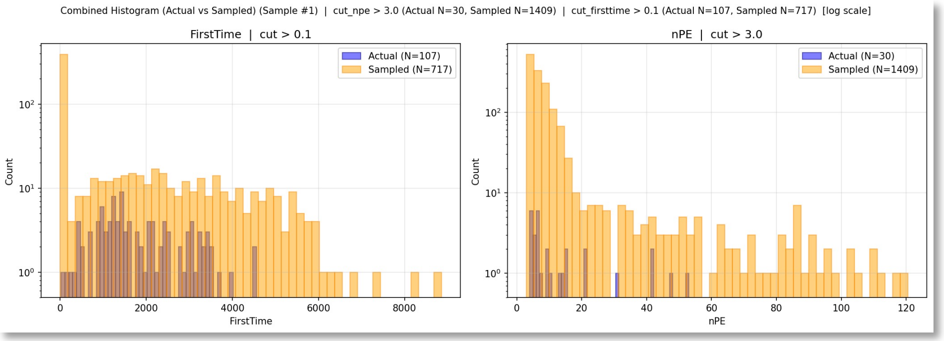
Bin1 : 66 ~ 100 PeV

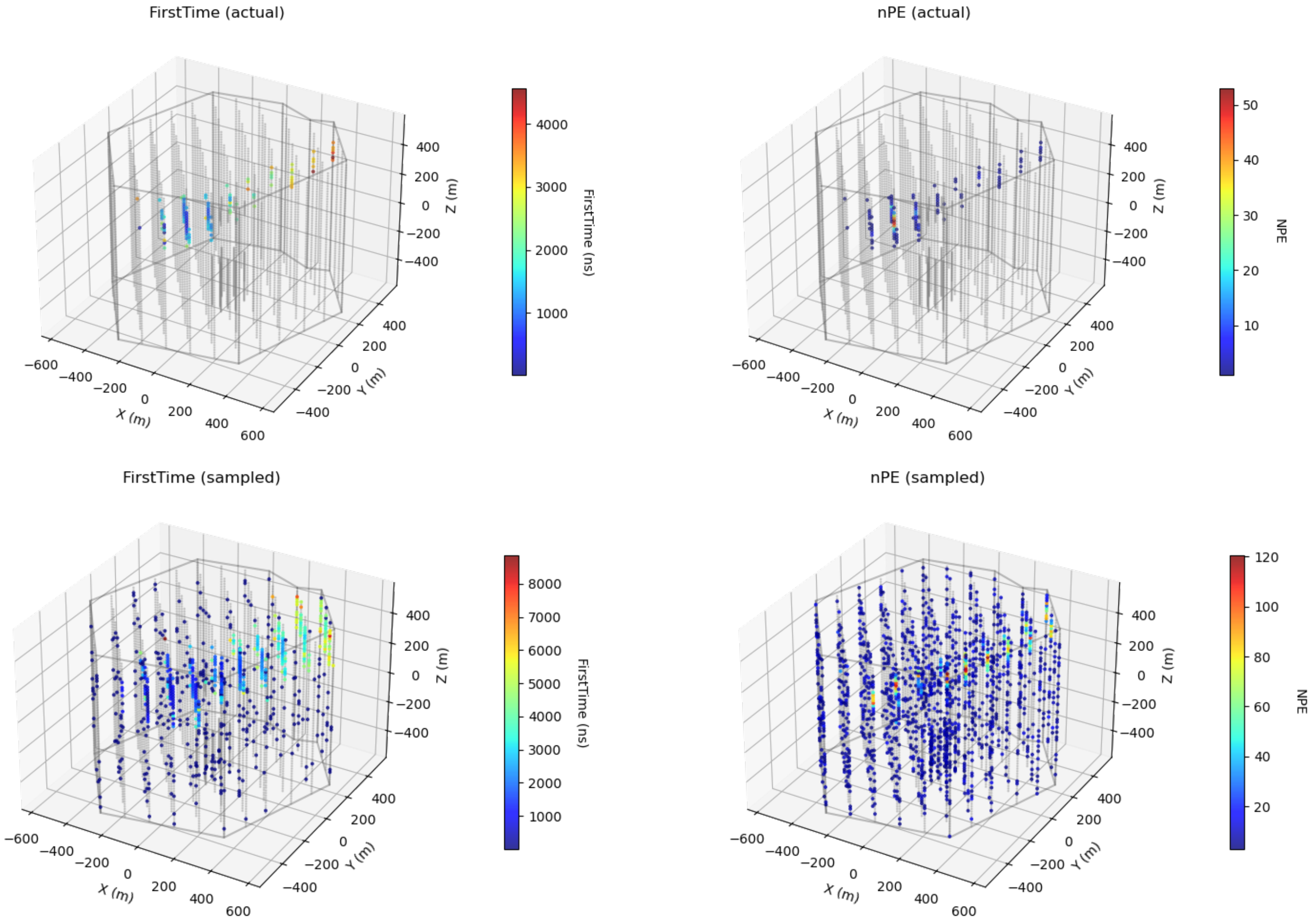
# Sample Performance



## Condition

- Energy : 70PeV
- Zenith : 94°
- Azimuth : 249°
- Vertex : (-9, -413, 283)



# 01 Bin 3 : 66~100PeV

# Sample



sample_cfg.py | CFG=2.0 | Sampled #1 | event 42333
Energy = 76.723 PeV, Zenith = 120.445°, Azimuth = 272.273°
Vertex Position (m) = (-256.14, -521.08, -258.55)